FYP-2 Mid Evaluation — 5-Minute Presentation Script

S26-017-R-GASR · FAST NUCES Islamabad · Supervisor: Dr Akhtar Jamil

Slides 1–7 are the talk. Slides 8–12 are the demo. Slides 13–24 are held for questions.

Member	Slides	Words	Time	Section
Ahmad Rashad 22i-1175	1 – 3	221	1:35 – 1:46	Goals and objectives · faculty feedback addressed
Shehroz 22i-0771	4 – 6	329	2:21 – 2:38	Progress: what we built, what we found, how it compares
Humair Khan 22i-2623	7	90	0:39 – 0:43	Remaining work, then drives the demo
Total		**640**	**4:34 – 5:07**	Counted from the spoken lines only. Fast end = 140 wpm, slow end = 125 wpm.

The rule this script follows about numbers. A number is only said aloud if it means something on its own — "all ten videos", "one camera instead of two". Numbers that only mean something next to another number — error scores, correlations, decibels — stay **on the slide**, where the panel can read them at their own pace, and are described in words instead. The slides still carry every figure, so "reporting adequate results" is satisfied; you just are not reciting them. If a panel member asks for a specific value, it is on screen or in the reference slides.

How to use this. These are the words, not notes to expand on. Do not read the slides out loud. **Do not stop for questions during slides 1–7** — the committee's five minutes are uninterrupted.

Ahmad Rashad — Slides 1 to 3 · about 1:35

Slide 1 — Title · 28 words · 12 s

Good morning. I'm Ahmad Rashad, with Shehroz and Humair. Our supervisor is Dr Akhtar Jamil. Project S26-017 — a research FYP on 4D surgical reconstruction and deformation prediction.

Slide 2 — What the project is for · 110 words · 47 s

A surgeon operating through one laparoscope sees a flat image of the present moment. Our goal is a system that reconstructs that scene in three dimensions and anticipates how the tissue will deform next — from that one camera, with no depth sensor and no annotation.

Point along the four cards. Say every status — the panel is marking progress.

Four objectives. Reconstruct in 3D from a single camera: complete, that was FYP-1. Learn how tissue

deforms, without labels: complete. Predict forward, with a sense of its own confidence: implemented. Evaluate it against the literature: in progress.

Pause. Read the question slowly — it is the sentence they should remember.

So the research question is this. Can future tissue geometry be predicted from ordinary surgical video — and can the system tell us when to trust it?

Slide 3 — Faculty feedback from FYP-1 · 83 words · 36 s

Say the feedback, say what was built. The file names are on the slide — do not read them.

We had four pieces of feedback after FYP-1, and all four are implemented.

Implement a base paper for comparative analysis — we implemented two, and Shehroz will come back to both of them.

Implement state-of-the-art methods — we built eleven of them, classical and learned, all measured the same way on the same data.

Do a detailed tuning and optimisation analysis — that is Module 8.

And begin a paper in conference format — there is an IEEE manuscript in the repository.

Step back from the laptop.

Shehroz.

Shehroz — Slides 4 to 6 · about 2:21

Slide 4 — The methodology, finalised · 90 words · 39 s

The methodology is now fixed. We track points on the tissue surface, look at the last few frames of how they moved, and predict where they go next. Nothing in that chain needs a human to label anything.

The model is a diffusion model — the family used for image generation, but here it generates motion rather than pictures.

The middle card. This is the one design decision worth naming.

One decision matters most. Rather than predicting the motion outright, it predicts the *correction* to a simple straight-line guess — so even when it learns little, it still beats guessing naively.

Slide 5 — Metrics chosen, and what they measured · 116 words · 50 s

Lead with the three questions, not the metric names. The figures are on the slide.

We measure three things: is the shape in the right place, is the motion pointing the right way, and does the model know when it is unsure.

The second matters because a shape score alone cannot tell a model that learned nothing from one that learned the right direction but the wrong distance. They score the same.

Three result cards, right side. Say what each means, not its value.

The answers. On raw accuracy we are not yet ahead of the simple baselines, and we lose on all ten videos — so it is not a fluke. But the direction is clearly learned, and the model's own confidence tracks its real error.

So the failure is in *how far*, not *which way* — a far more tractable problem.

Slide 6 — What we build on, and where we go past it · 123 words · 53 s

This is the comparison slide and it deserves its time. Four rows, left then right.

Four bodies of work sit next to ours.

EndoGaussian and Deform3DGS reconstruct surgical scenes beautifully — but with two cameras, and only ever describing what already happened. We reach comparable quality from one camera, then predict forward.

PointRNN is the closest published method to ours, and our main learned baseline. But it was built for rigid outdoor scenes, and gives one answer with no confidence.

Luo and Hu built the first diffusion model over 3D points — but it generates shapes, it does not forecast. Ours makes that idea conditional on time.

And surgical anticipation does predict the future — but in flat pixels, and only when told what the instrument will do.

Read the gold line as your closing sentence.

No published method does all of it at once.

Step back.

Humair.

Humair Khan — Slide 7 · about 0:39 , then the demonstration

Slide 7 — Remaining work until the final evaluation · 90 words · 39 s

Give each item a status, not a hope.

Five things remain, in the order they have to happen.

One: run the full comparison and the tuning studies. Both are written and tested — what is left is compute, not code.

Two: evaluate the two corrections we have already implemented. One of them needs no retraining at all.

Three: move to holding out an entire video, rather than frames from videos we also train on.

Four: fold the finished results into the manuscript.

Five: submit, and grow the dataset.

Go straight into the demo. Do not pause for questions.

And with that, let me show you the system running.

The Demonstration — Humair drives, slides 8 to 12

No verbatim script, because a live system will not run to one. These are the lines to have ready.

Slide 8 — while the notebook loads

Everything from here is live — nothing is pre-rendered. We load the trained model, check the data it is being tested on, predict a moment it has never seen, draw that prediction as an actual video frame, compare it against the real one, and then run past the end of the recording.

Slide 9 — while the render cell runs

There is a gap to bridge here. The prediction moves a few hundred tracked points; a picture needs the whole scene, which is tens of thousands. So the motion of the tracked points is carried outward onto everything around them.

Two honest notes about what you are seeing. Tissue moves only a pixel or two between frames, which is invisible, so the arrows are magnified — and that magnification is printed on the panel and never applied to any measurement. And the forecast runs past the end of the recording: the real-video panel stops and says so, while the predicted panel keeps going. That is the only output that shows prediction rather than describing it.

Slides 10 to 12 — describing the results

Agree beforehand whether you or Shehroz narrates these. One of you, not both.

The key finding: the model knows which way the tissue moves, not how far. Then uncertainty — when the model's own samples disagree with each other, its prediction really is worse, which means it can flag its own weak moments. A system that is confidently wrong is unusable in theatre; one that says "don't trust me here" is usable even when imperfect. And finally the full comparison against both simple baselines.

> **Demo checklist — do this the day before, not on the day.**
>
> - Runtime connected, GPU confirmed, checkpoint verified loadable.
> - Run the whole demo path end to end twice. Time the slowest cell and know what you will say over it.
> - Deck, notebook, checkpoint and outputs on **one laptop** — the committee requires it.
> - Fallback: if the runtime dies, show the already-executed notebook and say plainly that it is not re-executing. Do not pretend.
> - Screen resolution set so notebook text is readable from the back of the room.

Questions — and where the numbers live

Because the talk is deliberately light on figures, expect at least one question asking for them. That is fine: every figure is on a slide. Do not recite from memory — turn to the slide.

Topic	Answers · slides
Goals, scope, the model, why diffusion, training, overfitting, the architecture, the FYP-1 feedback	**Ahmad** · 14–16, 19

Methodology, metrics, baselines, quality tests, the comparison numbers, the direction finding, uncertainty	**Shehroz** · 5, 10–12, 17–18
Reconstruction quality against published figures	**Humair** · 24
The demo, rendering, the forecast, the dataset	**Humair** · 9, 21

The one answer to agree on. "Your model doesn't beat a simple baseline — why should this pass?" One person answers: *"Because we can tell you exactly why it doesn't. We measured that it learns which way tissue moves and fails only on how far; we tested and ruled out the two obvious alternative explanations; and we have implemented the two corrections that follow from that. We also get a confidence estimate that none of the baselines can give you. For a research project, a measured negative with an identified cause is worth more than an unexplained positive."*